

Know Who You Are: Earning Roles from Situated Peer Judgments

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Abstract

Multi-agent workflows often learn who should receive the next responsibility from acceptance signals or terminal success. Those signals are unsafe when a recipient must integrate a producer’s artifact: the same failure can arise from a producer defect, recipient-owned work, or a downstream component. We identify this failure as *responsibility confounding* and introduce *earning roles*, a protocol that publishes version-bound, typed recipient judgments as noisy observations while separately testing producer attribution. A later selector reads a legal snapshot, seals its assignment before execution, and updates only after a future outcome passes an independent reward check. This ordering yields testable noninterference, attribution-safety, and replay invariants. We define an ArtifactRole track for producer–recipient evidence and a complementary PeerSelect track for local selection and drift, with matched information, candidate menus, feedback delays, and complete-cost accounting. The resulting evaluation asks whether situated evidence improves future assignment quality beyond acceptance, terminal-only, and same-information contextual baselines, while measuring latency, forgetting, and false attribution. The scope is responsibility adaptation in dependency-bearing workflows, rather than universal expertise or autonomous workflow discovery.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; *Cooperation and coordination*.

Keywords

multi-agent systems, role formation, peer judgment, online adaptation, responsibility

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1 Introduction

In a dependency-bearing workflow, a collaborator’s value is observed through another agent’s attempt to use its artifact. Consider two queue-processing cases with the same rejection and failed team outcome. In one, a producer violates the priority-module contract; in the other, the module is correct but the recipient implements its consumer incorrectly. A terminal score or a bare rejection therefore points to the wrong future responsibility in at least one case. The

recipient is closest to the delivery, yet the same integration step can expose a producer defect or create recipient-owned work.

This is *responsibility confounding*: one feedback channel mixes producer correctness, recipient integration, and downstream outcome. Reputation and partner-selection systems make adaptive choice possible, but their usual interaction-level signal does not identify which side of a concrete producer–recipient dependency caused the failure [16, 19, 20, 22, 23]. Studies of cooperation among LLM agents likewise evaluate repeated-game or resource-sharing outcomes, rather than a version-bound artifact and its recipient-owned integration path [2, 15]. The problem is not solved by adding a larger score. The evidence must say what was observed, whose paths changed, and when that evidence became legal to use.

We ask: *when can the agent that actually receives and uses a delivery provide reliable evidence about the producer’s future responsibility?* The answer has a necessary sequence:

delivery → recipient use and judgment → typed observation
→ future assignment → held-out future quality and cost. (1)

A final score cannot identify which producer caused a defect. An acceptance bit cannot distinguish a useful artifact from an easy integration. A recipient’s normal integration work cannot become a producer penalty without an ownership check. Conversely, a role posterior that never changes a later pre-execution assignment is not role formation.

Our central operation publishes a completed, version-bound recipient judgment as a typed noisy observation, whether it reports acceptance, rework, or rejection. Producer-attributable evidence requires a separate structural ownership check; normal recipient integration cannot directly reward or penalize the producer. Publication does not silently update the policy. Before a later task starts, a legal owner reads the snapshot and the selector seals its assignment and propensity. Only after the later task has an independent outcome and a predeclared quality, responsibility, and cost rule passes can one delayed credit update occur. The protocol therefore separates what a recipient saw from what a future responsibility produced, while keeping the separation visible in a replayable ledger.

Figure 1 contrasts the two responsibility cases and shows how eligible evidence reaches a later sealed choice.

We call this protocol *earning roles*: a collaborator earns a future responsibility opportunity through attributable evidence that changes assignment under a fixed information and cost contract. We use *responsibility-aware role evidence* (RARE) for the protocol’s evidence semantics, not for a particular encoder, head, RLS update, or periodic refit. The primary claim is deliberately narrow: situated observation is useful only if it adds information beyond event counts and raw acceptance, terminal-only outcomes, and an equally informed contextual selector.

The paper makes four contributions.



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- We formulate responsibility confounding as an information and attribution problem, separating producer quality, recipient action, downstream adoption, final outcome, and UNKNOWN while preserving delivery, version, read cut, and propensity.
- We introduce the earning-roles protocol and three audit invariants: target outcomes cannot affect a sealed selection, recipient-only work cannot create producer reward, and replay cannot credit one assignment twice.
- We define an ArtifactRole track and a secondary PeerSelect mechanism track, with pre-registered root-level split requirements, same-information baselines, and complete resource accounting that distinguish evidence value from closed-loop team benefit.
- We make the claim falsifiable: no out-of-sample information, an equally informed contextual explanation, unsafe attribution, or a gain that disappears at matched cost invalidates the role-formation interpretation.

The following sections define the event and update order, comparison tracks, estimands, and falsification rules.

2 Related Work and Positioning

The nearest work differs from our question along three axes: what is observed, when state changes, and whose contribution receives credit. Table 1 makes that boundary explicit. Partner-selection and reputation methods motivate adaptive choice, while graph and role systems motivate changing organization; neither, by itself, identifies whether a concrete producer delivery caused a recipient’s action. Recent collaboration benchmarks measure team-level coordination and competition, while interactive agent benchmarks measure environment-level decisions and tool/user outcomes in stateful applications; neither by default turns a recipient’s integration trace into a producer-owned label [1, 14, 21, 26, 29]. A contextual-trust or delayed-bandit adapter supplies a strong same-information alternative, not a synonym for the proposed mechanism.

LLM collaboration frameworks enable role-based workflows, but task success does not resolve artifact ownership [4, 7, 24, 30]. Source judgments precede later assignments; target quality follows execution. Target outcomes may update future decisions but cannot revise a sealed assignment or label the source artifact without independent attribution. Bandit estimators are update comparators, while this protocol tests the timing, ownership, and service cost of that separation.

Cooperative multi-agent learning has long treated credit assignment as a separate problem, using counterfactual advantages or value factorisation to distribute a team payoff [5, 17]. Those methods establish why a shared outcome is insufficient for individual credit, but their reward/state contracts do not identify ownership of a concrete artifact inside a recipient’s integration path; this is the boundary tested here.

3 Problem Formulation

3.1 Objects and information boundary

At episode t , the task supplies context x_t and a legal candidate set C_t . The selector chooses one candidate before any candidate in C_t

executes:

$$a_t \sim \pi_\theta(\cdot \mid x_t, C_t, R_t), \quad a_t \in C_t, \quad (2)$$

where R_t is the public evidence snapshot available at read cut w_t . Only the selected candidate is charged for execution. The candidate emits a versioned delivery o_t with digest d_t ; a recipient j_t reads that delivery and takes action m_t . The recipient may accept, revise, reject, or be unable to complete the integration.

The smallest auditable event is

$$e_t = (x_t, C_t, a_t, o_t, j_t, J_t, m_t, Q_t, Y_t, \tau_t, c_t), \quad (3)$$

where J_t is the recipient judgment, Q_t is the producer-contract check, Y_t is an independent later or terminal outcome when the task defines one, τ_t is feedback-arrival metadata, and c_t is the complete cost record. The event also stores the candidate version, task/root identifier, artifact digest, feature-schema version, selection probability $p_t = \pi_\theta(a_t \mid x_t, C_t, R_t)$, read-cut digest, and evidence lineage.

Concrete instance. In the queue case, x_t is the JSON task request, $C_t = \{A, B\}$ is the legal producer menu, and $a_t = B$ selects producer B . The delivery o_t is B ’s versioned priority-module artifact with digest d_t ; recipient field $j_t = k$ records k ’s import, test, or repair action as m_t ; Q_t checks only the producer-owned queue and priority paths; Y_t is the later consumer outcome; and c_t contains model, tool, token, and wall-clock cost.

A repair to k ’s consumer path remains a publishable, typed hand-off observation if complete; it does not create producer-attributable evidence or reward for B .

The policy may consume only public events whose arrival is at or before its read cut. Hidden gold, a private scorer, a future target outcome, another policy’s state, and an unexecuted candidate cannot affect selection, retry, stopping, or update. A missing event is UNKNOWN; a complete but non-attributable event may remain a noisy observation, never a producer-negative label. This boundary is the difference between a role signal and a convenient retrospective score.

3.2 Observation and attribution gates

For a source episode, $O_t = 1$ requires a frozen contract and registry, sealed pre-action judgment, traceable delivery and recipient action, complete independent checks and terminal outcome, valid identifiers and digests, and completed source lineage. Acceptance, rework, and rejection have equal publication eligibility across their legally paired recipient actions; no arbitrary J -action mutation is assumed. Positive or negative checks do not by themselves filter publication. Before completion the judgment is auxiliary only; publication creates a versioned noisy observation and leaves persistent policy state unchanged. Separately, define $G_t = 1$ for producer attribution only when all of the following hold: (i) the frozen contract, candidate registry, changed paths, and independent producer check yield a unique producer structural owner (or a pre-registered source defect is uniquely bound to the producer); (ii) the recipient inspected o_t and its action is traceable to o_t ; (iii) recipient-owned integration paths are empty; (iv) $O_t = 1$; and (v) independent defect or quality evidence supports the producer-specific claim. The judge’s free-text target role is retained as a calibration observation and its disagreement with the structural owner is logged, but it cannot by itself create or cancel evidence. Otherwise $G_t = 0$: its

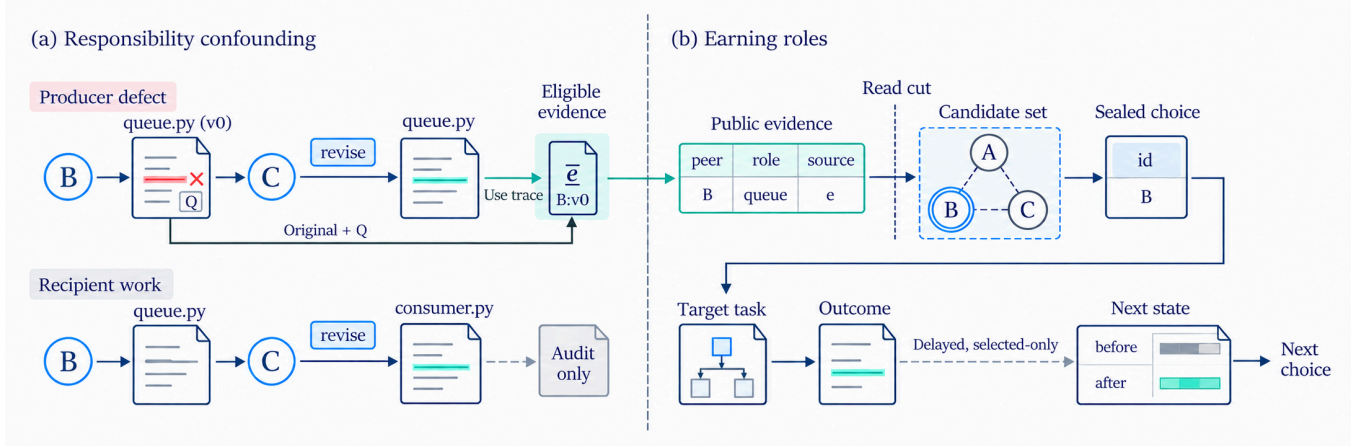


Figure 1: Responsibility confounding and earning roles. (a) Both cases receive a revise judgment. The original producer artifact is bound to B at version $v0$; its independent contract check Q and the recipient’s use trace support producer-attributable evidence e . Consumer-only changes remain an attribution-audit case. (b) At a legal read cut, public evidence informs a local choice that is sealed before target execution. After the checks in Figure 2, delayed selected-only feedback changes a future state. File changes, the sampled peer, and state highlights are schematic.

Representative method and problem	Observed signal and update	Responsibility / credit semantics	Experimental object	Difference / test here
Curiosity-Driven Partner Selection; Defection at First Sight [19, 23]	Interaction outcomes; partner state changes after social-game feedback	Peer-level interaction payoff; the cited formulations do not expose a producer-recipient artifact gate	Convention or dilemma games	Versioned delivery use and producer attribution
Reputation as a Solution to Cooperation Collapse [18]	Shared assessments and cooperation history; pooled reputation update	Broad reputation/cooperation score; the cited formulation does not expose producer and recipient ownership fields	Language-agent social cooperation	Public responsibility-safe evidence and explicit UNKNOWN
AgentNet; GPTSwarm [25, 30]	Agent traces, graph structure, or routing outcomes; organization changes over time	Route/role utility; the cited formulations do not define a producer-artifact credit label	Graph or workflow coordination	Recipient action bound to one delivery and future-owner assignment
Dynamic Role Assignment; MoRSE [13, 28]	Task context and role/subtask performance; roles or experts are reassigned	Role/expert utility; the cited formulations do not expose producer-artifact attribution inside recipient integration	Debate or mixture-of-experts workflows	Responsibility gate, cross-owner propagation, and delayed correction
Finite-time multi-armed bandit [3]	Selected-arm rewards; update after an observed arm payoff	Arm-level reward; no producer-recipient ownership contract	Pre-execution selection under observed payoff	Classical selected-reward control; contextual and delayed adapters require separate qualification

Table 1: Novelty boundary used in the paper. The rows are comparison families, not claims that each source implements the same task. A faithful adapter must preserve the listed visible information, arrival order, selected-only denominator, and complete cost before entering the scientific matrix.

typed observation can remain public, but it cannot create producer evidence or reward.

This gate distinguishes four observations that are often collapsed: a producer-content defect, a recipient’s own integration work, a sink or final-task failure, and an unresolved attribution. The public record retains the observation type and the reason for $G_t = 0$. A typed observation is not a native evidence identifier: the bridge to a future assignment requires its own qualified binding and is not yet established. This makes safety measurable: a method that refuses ambiguous evidence may have a higher UNKNOWN rate while producing fewer false producer penalties.

Figure 2 expands the separate interfaces for choosing at a legal read cut and updating after a target outcome.

4 Earning Roles: A Responsibility-Aware Protocol

The equations below specify the candidate realization used to qualify the protocol interfaces; they do not lock the final backbone or updater. Those choices remain an experimental question and will be fixed only after the same-information controls expose the relevant bottleneck.

4.1 Two-stage evidence and update

Publication and delayed updating are independent interfaces:

$$\text{publish}(e_t) \rightarrow \{\text{typed}, \text{UNKNOWN}, \text{invalid}\}, \quad (4)$$

$$\text{choose}(x_t, C_t, R_t) \rightarrow (a_t, p_t, \delta_t), \quad (5)$$

$$\text{validate}(j_t) \rightarrow \{\text{credit}, \text{UNKNOWN}, \text{invalid}\}, \quad (6)$$

$$\text{update}(\text{credit}) \rightarrow \{\text{updated once}, \text{no-op}\}. \quad (7)$$

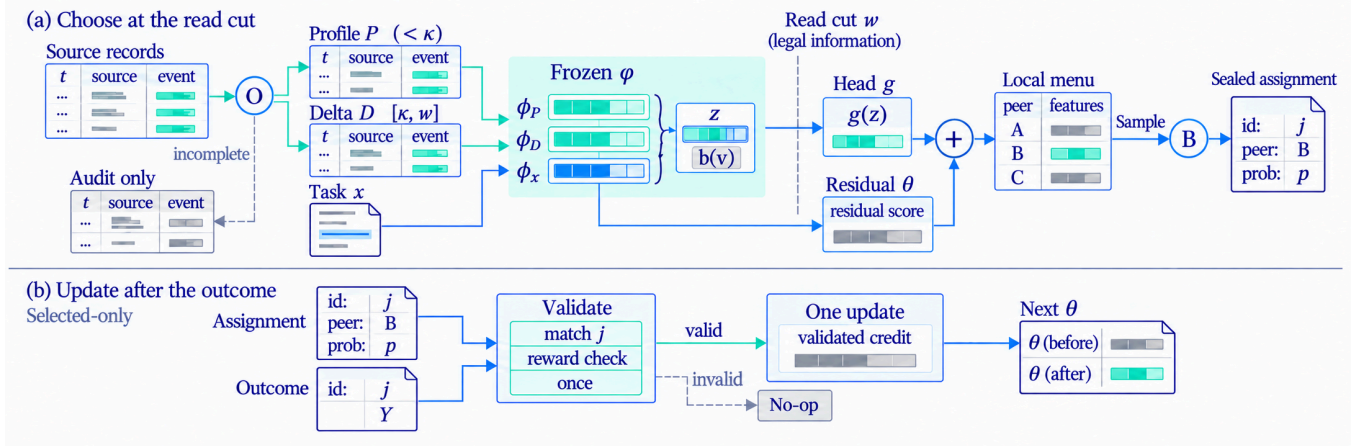


Figure 2: Choice and update interfaces of the candidate selector. (a) Complete source records ($O = 1$) are split at κ into profile P and delta D . Frozen task and history representations, with public metadata $b(v)$, feed head g ; the task representation also feeds a candidate-specific residual score. A local sample is sealed as (j, B, p) . (b) After target execution, that same assignment j and its independent outcome Y must pass identity, reward-contract, and once-only checks before changing the next residual state. The strips and sampled peer are illustrative; the encoder, head, and updater remain replaceable.

Publication may change a later assignment’s public input, but cannot update persistent policy state. For a later target assignment j , ledger binding is necessary but not a reward: credit is legal only if the assignment was written before target selection and task start, the selected candidate and propensity match the sealed preview, the target delivery and outcome are complete, a predeclared responsibility/quality/cost rule passes, and the assignment has not already been credited. The update then evaluates that future responsibility opportunity, never the source episode.

The event-time contract is therefore

$$\begin{aligned} \text{source selection} &\rightarrow \text{delivery} \rightarrow (Q, J, m, Y) \rightarrow O, G \\ &\rightarrow \text{evidence publication} \rightarrow \text{assignment} \rightarrow \text{target outcome} \quad (8) \\ &\rightarrow \text{delayed selected-only update.} \end{aligned}$$

A correction creates a superseding evidence version and affects only future snapshots. Duplicate, late, out-of-order, resource-failed, or invalid events are idempotent no-ops or explicit UNKNOWN records. Ledger replay must reproduce the same state digest and choice sequence.

4.2 Evidence representation and candidate score

A minimal public state for producer v in context r is

$$R(v, r) = (n(v, r), s(v, r), u(v, r)), \quad (9)$$

Here n counts typed observations, s summarizes noisy judgment categories without implying producer reward, and u audits unresolved attribution. Count-only or J -masked views remove judgment content. A candidate score can combine a task representation $h(x, v, r)$ with the public evidence term:

$$\text{score}(v \mid x, r) = w^\top h(x, v, r) + \beta \frac{s(v, r)}{n(v, r) + \alpha}. \quad (10)$$

For the queue instance, r is the priority-contract context, $h(x, B, r)$ is the fixed task-candidate representation for producer B , and n, s, u

summarize versioned observations and their attribution flags. A recipient-only repair can remain in n and u as a typed observation; it cannot supply producer reward for B .

The representation h , updater, and constants are replaceable implementation conditions rather than the scientific object. The score is a contract-level construction, not a theorem or a claim that one backbone is optimal. The selector uses a pre-registered exploration floor, records p_t , and uses preview-assignment-commit so that writing a future assignment cannot resample the selected candidate.

The same protocol yields three observable predictions. First, responsibility safety is measured by false producer-attribution and UNKNOWN rates under recipient-only interventions. Second, temporal validity is measured by read-cut and assignment-order invariants and by the absence of target-outcome fields at selection. Third, bounded online service is measured by per-event update latency, backlog, state bytes, and complete cost. These predictions map to the RQ2–RQ4 endpoints; they are not assumptions that the score automatically satisfies.

4.3 Executable selector and update contract

A candidate selector uses a frozen encoder for tasks, public profiles, and published observations. To avoid reusing a training label, it separates a timestamped profile from later evidence:

$$P_{v,k} = \text{summarize}\{e_i : O_i = 1, t_i < \kappa_k\}, \quad (11)$$

$$D_{v,k,t} = \text{summarize}\{e_i : O_i = 1, \kappa_k \leq t_i \leq w_t\}, \quad (12)$$

$$z_t(v) = [\phi(x_t) \parallel \phi(P_{v,k}) \parallel \phi(D_{v,k,t}) \parallel b(v)]. \quad (13)$$

Here κ_k is the profile cut, w_t the legal read cut, ϕ frozen, and $b(v)$ public version and uncertainty features. Empty inputs remain explicit; each event enters at most one of P , D , or target credit.

The candidate distribution is

$$q_t(v) = \text{softmax}_{v \in C_t} (g_\psi(z_t(v)) + \theta_v^\top \phi(x_t)), \quad a_t \sim q_t, \quad (14)$$

Table 2: State fields and their legal update event. A field is read only after its listed cut and never retroactively edits a sealed assignment.

Field	Source	Update boundary
Profile $P_{v,k}$	Published typed observations before κ_k	Next profile cut; immutable thereafter
Delta $D_{v,k,t}$	New typed observations after κ_k	Read watermark w_t
Residual θ_v	Validated selected target reward	Matching delayed update only
Uncertainty $u(v)$	Unknown, duplicate, or correction records	Monotone audit update
Assignment a_t, p_t	Candidate menu and snapshot	Sealed before target execution

Here g_ψ is a candidate fixed head and θ_v a residual updated only after a selected assignment passes the independent reward contract. Head, exploration, clipping, and learning rate are fixed before confirmation; profile-only, delta-only, residual-only, and no-history controls isolate components.

The candidate runtime contract requires five auditable operations. (1) It appends the sealed source judgment, publishes a complete typed observation, and checks attribution separately; incomplete records become UNKNOWN. (2) It advances a read watermark and materializes a state snapshot, recording the feature-schema and state digest. (3) It scores only the legal candidate menu and writes the candidate set, probabilities, sampled action, and assignment digest before the target starts. (4) When the target outcome arrives, it verifies the matching assignment, selected-only eligibility, version, idempotency key, and independent reward contract. (5) It applies one bounded residual update and emits a new state digest; profile and delta materialization wait for the next declared read cut. The sequence is required to permit replay after a crash without changing a sealed choice.

This candidate contract makes late judgments visible only to future snapshots, excludes recipient repair from producer reward, and leaves missing outcomes without updates. Confirmation must measure latency, state size, and utility; if a calibrated control matches it at equal information and cost, no fusion gain is claimed.

5 Benchmark and Baselines

5.1 Two complementary tracks

The primary track, **ArtifactRole**, is a TeamBench-derived candidate benchmark [10]. It is specified to expose producer artifacts, recipient dependencies, use/rework/rejection actions, responsibility checks, later assignments, downstream adoption, and complete cost. DIST1 queue-race and PIPE3 stream-processing are candidate structural roots; renamed seeds are not independent roots, and the development/confirmation split is frozen only by the versioned parity card before execution. Promotion requires material visibility, write permissions, scorer coverage, root independence, and clean replay. This stricter contract is intentional: TeamBench’s role separation supplies the dependency scaffold, while ArtifactRole adds the producer–recipient attribution and future-assignment fields needed by our question.

Producer conformance and terminal usability are scored separately: a recipient may normalize a nonconforming delivery into a

usable result. Every executed arm, including no-update, receives terminal evaluation. A prior valid, evidence-bound assignment and an independently validated reward are required for assignment credit, not for inclusion in the outcome denominator; native benchmark conformance remains a separate endpoint.

The secondary **PeerSelect** track uses a pinned local graph interaction benchmark for selected-only payoff, local candidate sets, online updates, drift, and service cost. It isolates update behavior but cannot establish situated artifact judgment or producer attribution. The tracks share event and selector interfaces while keeping tasks, labels, scorers, denominators, and claims separate.

5.2 Predeclared baselines

All primary arms receive the same candidate menu, initial opportunities, model/API/tool budget, exploration opportunity, feedback arrival schedule, failure handling, and complete cost accounting. Each arm reads only its legal history. Table 3 names the alternative explanations that the matrix must rule out.

The context-only Beta trust arm remains a diagnostic historical comparator; it is not the strongest same-information control. If the closest published method cannot preserve the same public fields, candidate menu, arrival order, selected-only rule, and complete-cost contract, the confirmation card records an explicit NO-GO; it does not replace that missing adapter with a self-authored weak control.

RLS, online logistic/SGD, periodic refitting, and any candidate incremental updater are training comparators. They are not counted as separate selection policies and are not presented as the innovation merely because they are fast. The contextual-trust-linear arm follows the same information contract as classical contextual bandits [12]; its delay handling is compared with the general delayed-feedback setting [8]. Agent ranking and human–LLM teaming provide additional positioning points, but neither substitutes for the producer–recipient label contract [11, 27].

6 Experimental Questions and Matrix

The matrix below is a pre-specified design scaffold, not an observed result table. A cell enters the confirmation analysis only after its root, stream, seed, budget, scorer, cost contract, stopping rule, and manifest digest are frozen before the corresponding calls; otherwise it remains planned or UNKNOWN.

6.1 Research questions and estimands

RQ1 (information value). Does an eligible situated judgment predict an independent producer-contract or later-use result beyond raw acceptance and terminal-only feedback? The primary estimands are out-of-sample log loss, Brier score, AUC when defined, calibration, and incremental predictive value on an independent root. The confirmation card also freezes a field-level feature manifest: a judgment-only view, a producer-check/terminal-only view, and their declared fusion. Artifact, structural owner, Qp, Y, read cut, arrival schedule, and cost remain fixed while legally paired recipient-judgment/action variants test whether the assignment or score actually responds to J.

RQ2 (assignment causality). Does public evidence change an assignment made before target execution, including when the future owner did not produce the source judgment? We measure

Arm	Read/update contract	Alternative explanation tested
Uniform	Candidate menu; fixed local-uniform sampling	Random allocation and exploration lower bound
No update	Initial prior and current context; feedback ignored	Static execution is sufficient
Raw acceptance	Recipient accept/reject only	Responsibility filtering adds no value
Terminal only	Independent final outcome only	Situated judgment adds no information
Contextual trust linear	Same public ϕ , menu, judgment, propensity, delay, and budget as RARE; ordinary feature-aware diagonal RLS	The effect is ordinary contextual trust or bandit learning
Pooled controller	Permitted public history under the same total capacity	A shared controller explains the gain
Closest published adapter	Faithful public-information adapter, if it passes qualification; otherwise an explicit no-go cell	Existing role/organization method is sufficient
RARE	Typed observations, attribution checks, and validated delayed credit	Candidate closed-loop mechanism

Table 3: Primary comparison boundary. A named arm enters the scientific matrix only after an executable adapter, information declaration, cost contract, and independent qualification pass.

assignment-change probability, propensity-weighted future quality, adoption, rework, and complete cost. Evidence from producer *B* cannot be consumed as if it belonged to producer *C*.

RQ3 (online behavior). Does the update path satisfy the service contract under selected-only delayed feedback? We report publish, read/assignment, and update p50/p95 latency, service lag, backlog, state bytes, CPU/GPU/RAM, token/tool/API calls, and replay time.

RQ4 (stability and boundaries). How does the method behave under task or peer drift, delayed or out-of-order feedback, version replacement, recipient-only work, mixed ownership, scorer failure, and resource failure? These are standard continual-learning and multi-agent non-stationarity stressors, but here they are coupled to responsibility attribution [6, 9]. We report old-root peak and mean forgetting, drift recovery window, UNKNOWN rate, correction consistency, and false producer attribution.

The primary unit of uncertainty is an independently reset stream. Episodes inside one stream are dependent. We use paired stream-level estimates, bootstrap or randomization intervals over streams, and pre-registered multiplicity control for the primary contrasts. Missing, invalid, or unknown events remain in the denominator and are never silently recoded as failures or successes.

Primary endpoint design. The design reserves four card identifiers: H1 uses independent-target Brier score (lower is better) for incremental situated signal; H2 uses future assignment quality-complete-cost utility (higher is better) against the same-information control; H3 uses selected-feedback update p95 latency and backlog under the fixed arrival budget (lower is better); and H4 uses false producer-attribution rate under ownership mutations (lower is better). Each card is required to fix its control, minimum practically important difference or precision target, exploration floor/positivity condition, stream-level 95% interval, and stopping rule before outcomes are observed; the current design-only manifest records which fields remain unfrozen.

Only these H1–H4 endpoints are primary for the headline interpretation. The remaining measures in the matrix are secondary or exploratory and cannot be promoted after outcomes are observed.

6.2 Complete experiment matrix

Table 4 is the main-paper overview of the execution skeleton. The detailed nine-cell cards expand each row into a root, split, policy, seed, budget, scorer, cost contract, and raw event manifest before confirmation; the overview does not replace those requirements.

Table 4 maps the two tracks to the four research-question endpoints while keeping the main-paper table readable.

Table 5 fixes the primary endpoint contract that will carry the headline comparison once the confirmation manifest is complete.

6.3 Budget, statistics, and reporting

Every model request reserves attempted calls, prompt tokens, maximum completion tokens, tool time, and storage before dispatch; errors release only unused reservations and retain their observed cost. We charge producer execution, recipient integration, judgment service, communication, retries, replay, verification, and returned artifacts. Development/search cost is reported separately from deployment cost. A pooled controller receives the same total capacity, and arms that do not use a judgment still pay the declared comparison cost or use a shared judgment call whose output is masked.

For each stream we record configuration, model route and version metadata, code and data hashes, seed, hardware, timestamps, raw API responses, token usage, intermediate actions, failure traces, UNKNOWN reasons, and per-sample outcomes. We report means and medians with 95% stream-level intervals, effect sizes for pre-registered contrasts, per-root heterogeneity, ITT and per-protocol denominators, and quality-cost Pareto points. The main endpoint is assignment-level future quality-complete-cost utility; source producer score is auxiliary and cannot replace it.

For every primary contrast, the experiment card fixes the direction, minimum practically important difference or precision target, stream-level 95% interval, and stopping rule before confirmation. A row without those fields is a design specification rather than an interpretable result.

6.4 Ablations and falsification

A count-only/*J*-masked control holds the same published hand-offs and costs while removing judgment-category information. The minimum ablation grid is: (A) no evidence/no update; (B) public evidence only, with persistent state unchanged; (C) delayed update

RQ	Track / unit	Contrast	Primary endpoint	Stress / split
RQ1	ArtifactRole / root	RARE / raw / terminal / contextual	Held-out Brier/log loss; calibration	Development → confirmation; matched public information
RQ2	ArtifactRole / target	Evidence+delayed credit / no evidence / delayed-only / pooled	Future quality–cost utility; assignment change	Pre-execution sealed assignment
RQ3	PeerSelect / stream	Updater / uniform / no-history / history-only	Payoff/regret; update p95; backlog/state	Selected-only; fixed call/token budget
RQ4	Both / mutation and drift	Ownership mutation / correction-replay / leave-one-out	False attribution; UNKNOWN; forgetting/recovery	Recipient/mixed; delayed order

Table 4: Pre-specified experiment matrix overview. Each row routes one research question to its track, contrast, primary endpoint, and stress condition. The confirmation cards expand these four rows into the nine execution cells and bind each cell to a root, split, policy, seed, budget, scorer, cost record, and raw event manifest before interpretation.

Table 5: Primary endpoint contract fixed before confirmation. Direction and unit are defined before outcomes are observed; numeric values and stream-level 95% intervals enter only after the matching card is executed.

Card	Primary endpoint	Direction	Unit
H1	Incremental situated signal	lower	held-out Brier / log loss
H2	Future responsibility allocation	higher	quality–cost utility
H3	Real-time service	lower	update p95 / backlog
H4	Attribution and stability	lower	false attribution / forgetting

only, without new public evidence; and (D) public evidence plus delayed update. Additional interventions remove responsibility filtering, selected-only restriction, delay, locality, correction/replay, or the assignment binding one at a time while holding information, model, candidate menu, and budget fixed. A content mutation that changes a visible judgment while holding terminal quality fixed must change the comparator score if the comparator actually consumes that judgment.

The main claim is withdrawn if an observation publication implicitly changes persistent policy; if a recipient repair is credited to a producer; if assignment or read-cut order is invalid; if unknown or duplicate events update state; if contextual trust explains the complete effect; or if gains disappear at matched complete cost. A null, a trade-off, or a qualified protocol result is scientifically valid, but it cannot be written as successful role formation.

6.5 Reproducibility and reporting

Before execution, each experiment card fixes the matrix cell, root, stream, model, budget, scorer, estimand, and stopping rule. A hash-linked ledger connects every aggregate to its raw calls and per-episode outcomes. Failed attempts retain their observed cost; incomplete cells remain TBD, never zero. We report safety without utility gain as a safety result, and prediction without changed future

assignment as an information result. Neither alone establishes role formation.

7 Results and Evidence Boundary

RQ1–RQ2 (information and assignment). The confirmation card will fill the held-out Brier score, matched-information comparison, assignment-change rate, and future quality–cost utility routed by Tables 5 and 4: TBD. Prediction without a sealed future assignment supports an information claim only; assignment without an independent target outcome does not establish responsibility transfer.

RQ3–RQ4 (service and safety). The latency, backlog, state-size, forgetting, UNKNOWN, and false-attribution cells remain TBD. We will interpret a safe null, a quality–cost trade-off, and a qualified positive result as different claims rather than as interchangeable evidence.

Current development traces expose attribution and measurement hazards, but do not establish predictive utility, future assignment gains, or real-time learning. No confirmation cell is filled from these traces. Results require qualified roots, independent streams, matched baselines, and complete cost.

8 Discussion and Limitations

The mechanism addresses responsibility opportunities in tasks with an actual producer–recipient dependency. It does not establish general expertise or autonomous workflow discovery. PeerSelect can isolate local adaptation, but cannot replace ArtifactRole’s attribution evidence; conversely, an ArtifactRole result cannot be interpreted as general bandit performance.

Judge error, selective exposure, root dependence, incomplete responsibility coverage, and judgment cost threaten validity. We therefore preserve UNKNOWN events, log selection propensities, assess judge reliability, and evaluate independent roots and streams. Terminal usability also differs from producer conformance: successful downstream normalization need not vindicate an upstream contract violation. Neither metric alone identifies the producer’s contribution to future team utility.

Backbone and updater selection remains an empirical question. Frozen representations with incremental heads, online logistic/RLS

updates, and periodic refitting are matched candidates. Latency alone cannot establish useful adaptation: a fast update may reinforce a misattributed label or forget previously useful distinctions. The drift and old-task evaluations must accompany update timing before a real-time learning claim is supported.

Deployment must expose refusals to learn, preserve correction lineage, and retain enough provenance to replay decisions after evidence compression. Tool selection can share this event interface, but requires its own execution oracle; peer judgments retain the ownership gate. The release will include versioned code, prompts, scorer contracts, raw-event manifests, and table-generation scripts. Offline replay reconstructs decision order; reproducing API latency and token costs requires fresh calls. These boundaries distinguish an executable protocol from demonstrated improvements in role formation.

9 Conclusion

This paper defines a precise object for self-evolving multi-agent roles: a recipient’s situated judgment becomes a typed noisy observation, producer attribution is checked separately, and a later sealed assignment is evaluated with independent quality and complete cost. The protocol makes each arrow auditable and gives close alternatives the same information and budget. The current draft records the method contract, benchmark and baseline matrix, statistical plan, and falsification boundary while leaving empirical cells empty. The decisive result will be whether the complete loop improves future responsibility allocation beyond equally informed alternatives without sacrificing attribution safety, online service, stability, or cost.

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